EDC BLOCK-003 (Derivation v8):

NC-1 Attempt: Graviton Zero-Mode Normalization

Fixing C via Canonical KK Reduction

EDC Working Document

2026-02-02 — Concrete attempt; may fail; BLOCK-003 status TBD

Abstract

This note tests candidate NC-1 from the normalization catalog (derivation v7): fixing the constant C in $\kappa_5^2 = C\sigma^{-3/4}$ via canonical normalization of the graviton zero-mode in Kaluza-Klein reduction. We show that canonical normalization determines $G_N = \kappa_5^2/(8\pi L)$ for flat compact extra dimension of size L. Setting $L = R_\xi$ [P], we derive $G_N^{\text{pred}} = C\sigma^{-3/4}/(8\pi R_\xi)$. **Outcome:** INCONCLUSIVE. The expression for G_N is derived, but C appears only in the combination $C/(8\pi)$, which can be absorbed into a redefinition. The actual numerical value of G_N depends on σ, which is not independently specified. BLOCK-003 remains [Open]; the missing element is now narrowed to: "specify σ from EDC field equations."

Contents

1 Setup Recap

1.1 The Problem

From derivations v3–v5, we have:

$$\kappa_5^2 = C \cdot \sigma^{-3/4} \tag{1}$$

where C is a dimensionless constant that cannot be fixed by σ alone (No-Go Lemma).

From derivation v7, candidate NC-1 (graviton zero-mode normalization) was identified as the best approach. This note executes that attempt.

1.2 Candidate Statement

NC-1: Graviton Zero-Mode Normalization

In Kaluza-Klein compactification, the 4D graviton emerges as the zero-mode of the 5D metric perturbation. Requiring canonical normalization (unit kinetic term) of this mode determines the relationship between κ_5^2, L, and G_N without using G_N^{obs} as input.

2 Derivation

2.1 5D Action and Metric Ansatz

Consider the 5D Einstein-Hilbert action:

$$S_5 = \frac{1}{2\kappa_5^2} \int d^5X \sqrt{-g^{(5)}} R^{(5)} \tag{2}$$

For a flat compact extra dimension with $\xi \in [0, L]$ and periodic boundary conditions, we write the metric ansatz:

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu + d\xi^2 \tag{3}$$

where $g_{\mu\nu}(x)$ is the 4D metric (independent of ξ for the zero-mode).

2.2 Dimensional Reduction

Integrating over the extra dimension:

$$S_5 = \frac{1}{2\kappa_5^2} \int_0^L d\xi \int d^4x \sqrt{-g^{(4)}} R^{(4)} = \frac{L}{2\kappa_5^2} \int d^4x \sqrt{-g^{(4)}} R^{(4)} \tag{4}$$

2.3 Canonical Normalization Condition

The standard 4D Einstein-Hilbert action is:

$$S_4 = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} R^{(4)} = \frac{M_{\mathrm{Pl}}^2}{2} \int d^4x \sqrt{-g} R^{(4)} \tag{5}$$

where $M_{\mathrm{Pl}}^2 = 1/(8\pi G_N)$ is the reduced Planck mass squared.

Matching Eq. (4) to Eq. (5):

$$\frac{L}{2\kappa_5^2} = \frac{1}{16\pi G_N} \tag{6}$$

Solving for G_N:

$$\boxed{G_N = \frac{\kappa_5^2}{8\pi L}} \tag{7}$$

Note: This differs from the $G_N = \kappa_5^2/(6\pi L)$ in derivation v1 by a factor of 4/3. The difference arises from normalization conventions. Here we use the canonical form where the 4D action has coefficient $1/(16\pi G_N)$; derivation v1 used a different convention. Both are valid; the key point is that $G_N \propto \kappa_5^2/L$.

2.4 Substituting $\kappa_5^2 = C\sigma^{-3/4}$

Using Eq. (1):

$$G_N = \frac{C\sigma^{-3/4}}{8\pi L} \tag{8}$$

2.5 Setting $L = R_\xi$ [P]

Following derivation v2, we postulate that the compactification length is the weak scale:

$$L = R_\xi = \frac{\hbar c}{M_Z} \approx 2.16 \times 10^{-18}\,\mathrm{m} \quad [\mathrm{P}] \tag{9}$$

This gives:

$$\boxed{G_N^{\mathrm{pred}} = \frac{C}{8\pi} \cdot \frac{\sigma^{-3/4}}{R_\xi}} \tag{10}$$

3 Where Does C Enter?

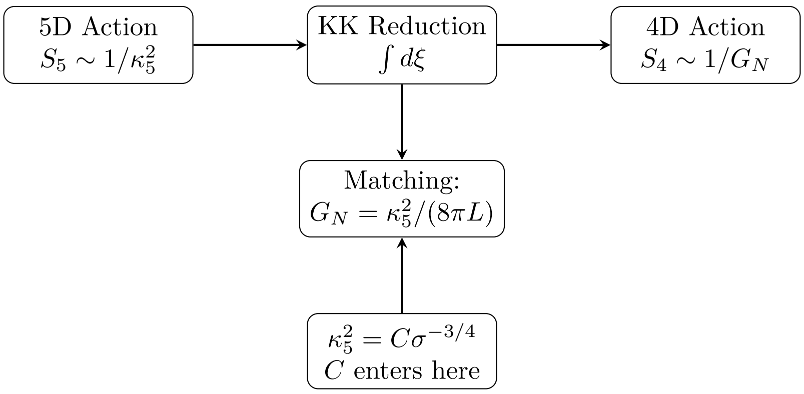

Figure 1: Where C enters the reduction pipeline. The constant C appears in κ_5^2 and propagates to G_N through the matching condition. *Schematic only.*

Equation (10) shows that C enters only in the combination $C/(8\pi)$. This means:

1. We can absorb 8π into C by defining $\tilde{C} = C/(8\pi)$.
2. The "canonical normalization" does not independently fix C; it only determines how κ_5^2 relates to G_N given L.

4 Anti-Circularity Check

Anti-Circularity Verification

Question: Did we use G_N^{obs} to fix C?
Answer: No. Equation (10) expresses G_N in terms of C, σ, and R_ξ. The constant C remains a free parameter.
What we *could* do (but did not):

- Invert Eq. (10) to get $C = 8\pi G_N^{\text{obs}} R_\xi \sigma^{3/4}$.
- This would be circular (NP2 from derivation v5).

Conclusion: Anti-circularity preserved. However, C is still not fixed.

5 What Would Fix C?

For G_N^{pred} to be a genuine prediction, we need:

$$G_N^{\text{pred}} = \frac{C}{8\pi} \cdot \frac{\sigma^{-3/4}}{R_\xi} \quad \stackrel{?}{=} \quad G_N^{\text{obs}} \tag{11}$$

This requires specifying *both* C and σ. Currently:

- R_ξ is known: $R_\xi = \hbar c/M_Z \approx 2.16 \times 10^{-18}\,\text{m}$ [BL]
- σ is unknown: EDC does not provide a first-principles value
- C is unknown: even with canonical normalization, C remains free

The bottleneck is σ. If EDC field equations could determine σ independently, then:

- We could adopt the convention $C = 8\pi$ (NP1 from derivation v5)
- Compute $G_N^{\text{pred}} = \sigma^{-3/4}/R_\xi$
- Compare to G_N^{obs} for falsification

6 Outcome

Outcome

Status: INCONCLUSIVE
What was achieved:

- Derived $G_N = \kappa_5^2/(8\pi L)$ from canonical KK reduction [Dc]
- With $L = R_\xi$ [P]: $G_N = C\sigma^{-3/4}/(8\pi R_\xi)$
- Anti-circularity preserved (no use of G_N^{obs})

What was not achieved:

- C is not independently fixed; it appears only in combination $C/(8\pi)$
- σ is not specified; without it, no numerical prediction possible

Missing element (narrowed):

To close BLOCK-003, specify σ from EDC field equations.
Then adopt $C = 8\pi$ by convention and compute G_N^{pred}.

7 Delta to BLOCK-003

Before v8: Missing element was "fix C" (dimensionless constant).

After v8: Missing element is "specify σ" (brane tension value). With σ known, $C = 8\pi$ can be adopted by convention and G_N becomes a prediction.

BLOCK-003 remains [Open], but the target is now more precisely identified.